



Wormhole tunnels possible?

According to research, there may be ways to create wormholes predicted by general relativity. <http://dgit.in/jun21-59>



Sleep evolved before brains

Some of nature's simplest animals suggest that sleep evolved long before centralized nervous systems. <http://dgit.in/jun21-60>

is why you need to swing the arms at the right time too. If people try to slow down from a resonant walk, they actually find that they will not be using less energy if they slow down any more. The resonant walk is achieved at the moment when it becomes harder to slow down. Researchers from the Vrije Universiteit in Amsterdam (VU) applied the principle to dinosaurs, using the world famous T-Rex, to see how the creature could achieve a resonant walk. Here, the tail of the animal and its movement was expected to be an important factor in helping the creature resonate. For the purpose of the study, the researchers built a 3D model of Trix, the T-Rex specimen at the Dutch National Museum of Natural History, then added digital muscles and ligaments for the biomechanical analyses. They found that the tail of the animal behaved much like a suspension bridge, with the tail swinging up and down, and left and right as well with every step taken by the T-Rex. This is one of the first studies of its kind taking a crack at figuring out how the tails of the dinosaurs moved when it was walking, and found that the resonant walking speed for Trix was around 4.6 km/h. This meant that if a human would ever have had a pet T-Rex, they could easily walk side by side, much like walking a dog.

Another way of looking at this approach of increasingly refining the constraints on how dinosaurs moved, is finding out all the ways that they did not, or could not, and exclude those. As you can guess, some species are more well studied

than others, and this is where the approach of excluding the types of movements can provide insights into the possible ways in which the lesser studied species moved. Traditionally, this involved just eliminating the ways in which the fossilised bones of dinosaurs could possibly fit together to form the whole animal. Get it wrong at the skeletal level, and the bones can strike each other while moving, or the animal can just come apart. However, apart from just the bones, there are muscles, tissues and tendons to account for as well, and doing this has been far more difficult with researchers using a range of approaches including musculoskeletal models, trackway measurements and simulations with robots, all of which have their limitations. Researchers from Brown University have improved the approach for biomechanical evaluation of extinct animals by developing a new imaging technique known as X-ray Reconstruction of Moving Morphology or XROMM. The approach can potentially be used for biomechanical analysis of any extinct species, but in the proof of concept study the researchers looked at dinosaurs, and two living relatives that have similar movement, the American alligator and the helmeted guineafowl. The approach involved capturing X-Ray videos of a range of movements in the two creatures, with the limbs being manipulated in different ways, as well as CT scans of the skeletons. These were used to gather

data of 600,000 different joint poses, allowing the researchers to create what are known as mobility maps. The researchers were able to pry apart patterns in the data that related the joint mobility to specific poses used by both alligators and birds.

Armita Manafzadeh, a Ph.D. candidate in ecology and evolutionary biology at Brown and lead author of the research says, "By combining the



XROMM images of birds on top and alligators on the bottom

latest technology for studying joint motion with unprecedented amounts of joint pose data, we've uncovered surprising new information that will improve reconstructions of locomotion in extinct animals." The success of the study was that the joint mobility data, when churned using computers, actually provide palaeontologists with much more information, based on the information that is already available. Manafzadeh explains, "Thanks to these pose usage patterns, it turns out that what dinosaurs couldn't do with their joints will give us some big clues about how they walked. What's more, this information has been right in front of us, hidden within the data that paleontologists are already collecting. We just didn't know how to appreciate it yet."

There is still more to be known about these patterns though, the researchers next hope to investigate how these patterns evolved in the birds, alligators and dinosaurs in the first place, and why. The researchers also want to investigate how broad these patterns are. The technique and the tools for figuring out the locomotion of extinct animals has been made available to the scientific community. 



The two legged crocodile, so you do not have to imagine it